

Supplement: Semantic Structure and Revision in Temporal Evidence Graphs for Generative Monitoring

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S1 Construction and Projection

Table S1: Complete stored node vocabulary. Nodes have a type, not an edge domain/range.

Type / label	Meaning	Creation and provenance	Status
Subject	Dataset participant identity	with_entities: dataset/subject; record scope	administrative
Sensor	Session channel identity	with_entities: observation metadata. channel; recording channel	administrative
Observation	Hashed recording window and sampling coordinates	adapter/extractor; recording hash and sample range	deterministic
Feature Version	Numerical feature or explicit availability watch	extractor or watch subscription; source_ids, extractor and units	deterministic
Claim Version	Generated bounded proposition and declared inputs	displayed_records; accepted flag for checked methods; raw candidate, query, claim fields	proposed/ checked
Explanation Version	Immutable displayed text and member claims	generation or maintenance recomposition; case_id, previous version, withdrawn IDs	administrative
Review Event	Recorded reviewer action	isolated dashboard action; action metadata and source_ids	administrative
Support Assessment	Support outcome at a knowledge time	GraphStore.assess_many; record ID, known_at, trigger_id, state, value	validated outcome

Type / label	Meaning	Creation and provenance	Status
Record (no payload)	Unresolved referenced ID, if present	GraphStore MERGE of source endpoint; referencing record	administrative/unresolved

Table S2: Complete relation vocabulary. R denotes any Record kind; P an unresolved placeholder. Assessment is not a Record kind.

Relation	Domain $\rightarrow$ range	Meaning	Creation / status
DEPENDS_ON	$R \rightarrow R/P$	Dependent to declared input; does not certify necessity	record.source_ids; deterministic or model proposal
DERIVED_FROM	feature $\rightarrow R/P$	Feature extraction provenance	feature.source_ids; deterministic
SUPPORTS	$R/P \rightarrow$ claim	Accepted-at-insertion citation/prerequisite mirror	accepted claim. source_ids; checked at insertion; not perpetual support
FOR_SUBJECT	$R \rightarrow$ subject	Scope membership, not reasoning	with_entities and record_relations; administrative
OBSERVED_BY	observation $\rightarrow$ sensor	Recording channel association	observation. metadata.channel; deterministic
SUPERSEDES	$R \rightarrow R/P$	New immutable version to previous ID	supersedes_id; administrative versioning
CONTAINS	explanation $\rightarrow R/P$	Display membership	explanation.source_ids; administrative
REVIEWED_BY_EVENT	$R/P \rightarrow$ review	Item acted on by review event	review.source_ids; administrative
CONTRADICTS	$R \rightarrow R$	Revision triggered a contradicted assessment	assess_many(state=contradicted); validated outcome; time in assessment
ASSESSED_AS	$R \rightarrow$ assessment	Record has a time-indexed support assessment	GraphStore.assess_many; validated outcome

The node table covers all seven Record kinds plus adapter-created assessments and unresolved reference placeholders. The latter are possible through source-endpoint MERGE operations but have zero observed count in the exported original scopes. Domains describe adapter behavior, not a narrower schema that the database does not enforce. In particular, source IDs produce broad Record-to-Record relations; admission checks supply the numerical and task restrictions. Identity

constraints enforce scope/ID uniqueness. Acyclicity and immutable-record checks are implemented in Python; neither OWL entailment nor SHACL validation is claimed.

The conceptual PROV-DM mapping treats observation and feature versions as entities, feature extraction and generation as activities recorded in metadata, and source/version links as derivation and revision information. The actual database does not materialize a complete PROV activity vocabulary. A `SupportAssessment` is an outcome keyed by scope, record ID, knowledge time and trigger ID. The SQLite control stores equivalent assessment rows without Neo4j's extra assessment nodes. `CONTRADICTS` has no edge timestamp, so its associated assessment supplies time. Raw rejected proposals are archived separately from persisted `ClaimVersions`.

An audit finds 432 record/time groups with multiple assessments, including 245 with different states across the original exported scopes. Sequential triggers can share an ingestion timestamp without an explicit within-time sequence field. The semantic projection preserves every tied outcome and marks conflicting summaries ambiguous, rather than treating database element order as chronology. Independent proposition scores and display membership use the event log and saved explanation versions, so this audit changes neither the original outcomes nor the new grounding/recall results.

The stored graph contains immutable versions, explanation containers, subject and sensor identities, review events and assessments. The active evidence-and-claim graph selects latest visible logical feature/observation versions, current explanation members and their claim ancestors. Original observations required by visible feature provenance are also retained. The query core starts from a specified explanation, closes over declared claim parents, and retains cited features resolved according to the query's temporal mode. Identity, container and assessment objects become sidecars. Observation provenance is a separate expansion. Claims' asserted intervals come from their proposition metadata, not from the query-interval wrapper used by the runtime.

A citation remapped from an old immutable feature to its current logical version is exported as `CURRENT_CITATION_PATH` with the original citation and version-path witness IDs. It is never relabelled a directly stored Neo4j relationship. Degree/depth calculations exclude `SUPPORTS` mirrors and count each dependency edge once. Full storage counts immutable objects; core statistics count visible propositions/versions per snapshot. Snapshots and replay copies remain clustered within their source episode and participant.

The representative graph rule takes the lexicographically first median-size eligible correction core per real source. Eligibility requires visible claims and a revised cited feature. A separately registered maximum actual-impact rule selects the same case in both sources, so no second purportedly distinct example is drawn. The exports retain both rule labels and the complete eligible lists. WESAD S11 and PPG-DaLiA S14 are therefore reproducible examples, not selected because they show a propagation advantage. Both show direct operand citations.

S2 Independent Semantics and Extended Results

Table S3. Node counts for full visible storage, active evidence-and-claim graphs and query cores, with subject-cluster 95% intervals. Core/stored is a percentage; final column counts nonzero exposure among eligible revision batches. Ten subjects and 20 episodes per source.

Source	Stored		Active		Core	Core/stored (%)	$\xi > 0$
Synthetic	184.3	[183.6, 185.0]	170.3	[169.8, 170.7]	2.8	[2.3, 3.3]	1.5 [1.3, 1.8] 0/66
WESAD	185.6	[184.3, 186.8]	171.6	[170.4, 172.6]	4.1	[3.0, 5.2]	2.2 [1.6, 2.8] 0/68
PPG-DaLiA	185.5	[184.2, 186.8]	171.4	[170.3, 172.6]	4.0	[2.9, 5.2]	2.1 [1.5, 2.8] 0/76

The evaluator reads the observation/revision log and task specification; it never imports graph support assessments or auxiliary-verifier decisions as reference truth. Canonical identities include source, participant/session, interval, quantity, operation, normalized unit/value, temporal mode and resolved logical versions. Numerical matching preserves the original tolerance. Required slots are matched once; raw duplicate propositions remain in the precision denominator. Fully unanswerable requirements contribute NA recall, while partially answerable tasks contribute only answerable slots. An empty answer to an answerable question has zero recall. Subject means are computed after episode aggregation; pooled counts are supplied for interpretation and are not the same estimand.

Table S4. New semantic protocol versus the unchanged B0 contract outcomes. Grounded background is a subset of grounded formerly invalid content. The remaining disagreements include scope/dependency and citation admission restrictions; no old valid claim becomes ungrounded.

Source	Old invalid	Grounded now	Of these: background	Valid now false
Synthetic	254	178	107	0
WESAD	278	162	115	0
PPG-DaLiA	267	159	111	0

The machine-readable proposition ledger preserves each raw sentence, canonical proposition, admissible numerical witnesses, match to requirements and limited prose audit. That audit flags literal discrepancies and interpretation terms, not every possible semantic implication. The new structural audit enumerates all assignments to four or eight primitive tests and compares each proposed witness with the task-defined proposition. Distinct logically equivalent derivations are allowed. Soundness of each witness and completeness of the alternative family are separate: a sufficient route can be sound while omitting another valid route. True initial Boolean values alone do not establish either property.

Table S5. Independent competency checks on displayed M1 waveform claims. Each eligible comparison must expose both operands; known-version and source-window answers are checked against immutable events. Alternative-support behavior is tested separately in the structural experiment.

Source	Competency question	Correct / eligible
Synthetic	source_window	336/336
Synthetic	known_version	336/336
Synthetic	comparison_witness	13/13
WESAD	source_window	440/440
WESAD	known_version	440/440
WESAD	comparison_witness	3/3
PPG-DaLiA	source_window	425/425
PPG-DaLiA	known_version	425/425
PPG-DaLiA	comparison_witness	2/2

Table S6. Event-aligned M1 prior-answer checkpoints across all variants. Each source/position has 160 transitions from 20 episodes and ten subjects. Intervals are subject-cluster 95% intervals. No continuous-time loss or finite recovery time is inferred from these points.

Source	Position	P [CI]	R [CI]	Grounded / all	Matched / req.
PPG-DaLiA	arrival	57.9 [46.7, 69.7]	41.2 [29.4, 54.2]	191/271	122/290
PPG-DaLiA	before	100.0 [100.0, 100.0]	75.8 [65.0, 87.5]	271/271	202/260
PPG-DaLiA	maintained	100.0 [100.0, 100.0]	41.2 [29.4, 54.2]	191/191	122/290
Synthetic	arrival	62.0 [55.7, 68.1]	46.5 [36.7, 57.4]	138/211	137/290
Synthetic	before	100.0 [100.0, 100.0]	79.2 [67.1, 91.2]	211/211	210/260
Synthetic	maintained	100.0 [100.0, 100.0]	46.5 [36.7, 57.4]	138/138	137/290
WESAD	arrival	64.2 [54.1, 73.2]	44.4 [32.5, 55.5]	207/281	131/290
WESAD	before	100.0 [100.0, 100.0]	76.2 [65.0, 87.5]	281/281	203/260
WESAD	maintained	100.0 [100.0, 100.0]	44.4 [32.5, 55.5]	207/207	131/290

Table S7. Structural candidate semantics: initial propositions are true in all cases; witness soundness and answer-root coverage with an admissible witness. Entries are percentages with subject-cluster intervals; each source/stage has 80 cases from ten reused episodes and five subjects. Accepted/displayed refers to original extension admission. Primitive test names are normalized for independent proposition truth, not silently admitted.

Source	Stage	Claims	Witness % [CI]	Root % [CI]
PPG-DaLiA	Admitted	250	100.0 [100.0, 100.0]	87.5 [83.8, 91.2]
PPG-DaLiA	Final	336	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
PPG-DaLiA	Raw	336	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
Synthetic	Admitted	246	100.0 [100.0, 100.0]	87.5 [83.8, 91.2]
Synthetic	Final	344	92.1 [84.0, 100.0]	95.0 [90.0, 100.0]
Synthetic	Raw	344	92.1 [84.0, 100.0]	95.0 [90.0, 100.0]
WESAD	Admitted	245	100.0 [100.0, 100.0]	86.2 [83.8, 87.5]
WESAD	Final	336	77.0 [69.4, 83.5]	86.2 [80.0, 91.2]
WESAD	Raw	336	77.0 [69.4, 83.5]	86.2 [80.0, 91.2]

S3 Structural Registration, Admission Correction and Cost

The initial protocol, deterministic selection and source hashes precede new inference. Development pilot v1 produced terse test-token sentences and direct roots. A second prospectively documented pilot clarified natural-language sentences and reuse of earlier claim IDs, with a generic unrelated grouping example. No required model links were compiled in. Both pilots use development episodes; the core freeze follows pilot v2. A separate pilot-amendment manifest records the added development budget and prompt hash. The core allows at most one repair per case and no post-correction generation.

Table S8. Separate extension inference accounting. Roots count candidates admitted by that stage’s runtime. All calls use Qwen3-8B BF16 on the existing RTX 5090. CPU database replays add no model requests. The core’s 325 calls remain below its 480-call cap.

Stage	Cases	Calls	Repairs	Roots	Minutes	Input tokens	Output tokens
pilot v1	12	21	9	12	0.42	23,729	5,437
pilot v2	12	19	7	9	1.56	32,774	22,351
core	240	325	85	209	17.23	476,646	282,095

The frozen extension’s proposition-name table omitted singleton primitive names although their comparators were explicit in the input. The independent scorer resolves such names to those definitions. A separately scoped corrected replication additionally makes them admissible to the unchanged validator by extending the proposition map. It preserves the final candidate JSON, source events, witnesses, parent IDs, update conditions and repair opportunities. Candidate hashes connect the corrected run to the original. It makes no new GPU calls. Its 229 admitted roots replace neither the original 209 admission result nor any minimum-v1 result.

Table S9. Separate primitive-name normalization replication: 240 cases, four update conditions and five methods (4,800 cells). All original model outputs and parent links are unchanged. Correction intervals use 15 dataset-scoped participants and 30 episodes.

Method	Required	Missed	Collateral	Corrected % [CI]
B0	1294	1294	0	0.0 [0.0, 0.0]
B1	1294	208	0	83.9 [83.0, 84.7]
B2	1294	208	0	83.9 [83.0, 84.7]
M1	1294	0	0	100.0 [100.0, 100.0]
B3	1294	0	0	100.0 [100.0, 100.0]

The normalized replication still realizes only evidence-to-answer depths one and two. Undefined or malformed references are retained as failures; they are

never fixed by inserting edges. Its paired database outcomes agree, its eligible direct-miss sets match exposure exactly, and its admitted two-route answers retain support after primary-route loss. Full candidate/decision/replay records and the independent audit are distributed alongside the primary run.

The structural replay timing fields include both direct and transitive reach queries for diagnostic comparison, current snapshot retrieval, and evaluation of the whole small program before scheduled writes. They are useful execution accounting but not optimized affected-subgraph latency. Source-level numerical corrections, benign changes and irrelevant changes are represented explicitly. No price-per-hour claim is inferred because a billing tariff was not recorded.

S4 Preserved Minimum-v1 Accounting

This section retains the completed study’s definitions, counts and strict contract interpretations. These are not retrospectively replaced by the new semantic scores. The archived original main paper and report provide their full narrative. Independent generation in the original five methods means conditional correction denominators can differ; the new extension removes that ambiguity by replaying one common candidate. The auxiliary verifier is a model audit, not a gold oracle.

Table S10. Original source and prepared-record inventory.

Source	Dev/test subjects	Test episodes	Scenarios	Cases	Raw values
Synthetic	10/10	20	80	1,200	1,900,800
WESAD	5/10	20	80	1,200	6,220,800
PPG-DaLiA	5/10	20	80	1,200	6,220,800

Table S11. Original method ablations and maintenance behavior.

ID	Storage	Checks	Maintenance	Repair allowance
B0	Indexed records	No	No	No
B1	Indexed records	Yes	Direct	One
B2	Neo4j	Yes	Direct	One
M1	Neo4j	Yes	Full	One
B3	SQLite + recursion	Yes	Full	One

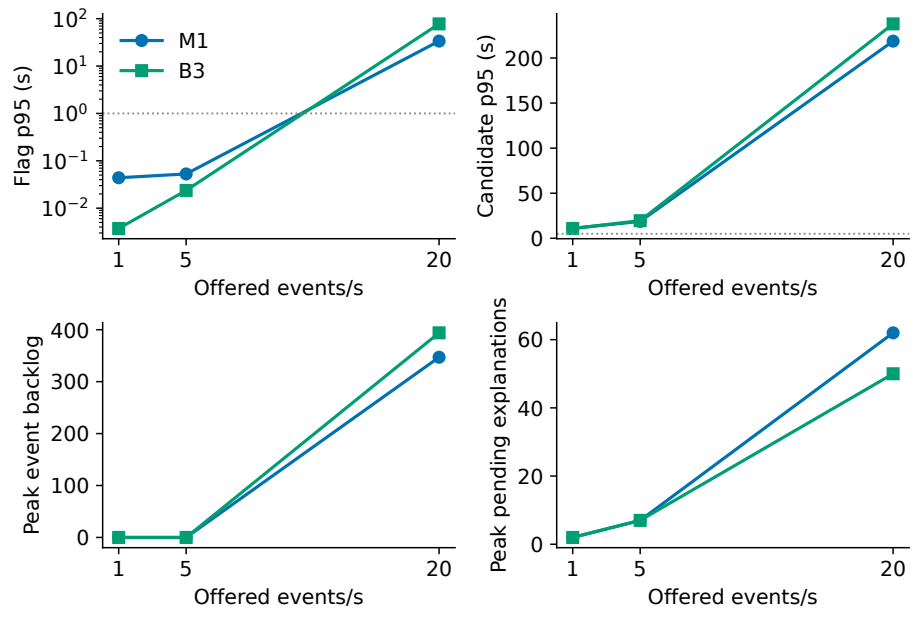


Fig.S1. Preserved four-panel systems measurements. Finite generation latency is candidate-completion latency. No candidate satisfies both required facts, so these curves do not measure successful correction time.

Table S12. Frozen exact-contract reliability and answer coverage. Correctly labelled off-target background remains a contract error in this table.

Source	Method	Case err.%	Claim err.%	Recall%	Coverage%	Claims
Synthetic	B0	52.9	43.2	78.8	60.0	588
Synthetic	B1	0.0	0.0	79.3	60.5	335
Synthetic	B2	0.0	0.0	79.3	60.5	336
Synthetic	M1	0.0	0.0	79.3	60.5	336
Synthetic	B3	0.0	0.0	79.3	60.5	335
WESAD	B0	59.2	39.2	76.0	54.1	710
WESAD	B1	0.0	0.0	76.9	55.9	440
WESAD	B2	0.0	0.0	76.9	55.9	440
WESAD	M1	0.0	0.0	76.9	55.9	440
WESAD	B3	0.0	0.0	76.7	55.5	439
PPG-DaLiA	B0	56.2	38.5	76.9	55.9	693
PPG-DaLiA	B1	0.0	0.0	76.9	55.9	425
PPG-DaLiA	B2	0.0	0.0	77.1	56.4	427
PPG-DaLiA	M1	0.0	0.0	76.7	55.5	425
PPG-DaLiA	B3	0.0	0.0	76.9	55.9	426

Table S13. Original required-change correction and collateral effects, conditional on each method's displayed claims.

Source	Method	Corrected/ required	Complete (%)	Residual	Collateral/ unaffected
Synthetic	B0	0/71	0.0	71	0/245
Synthetic	B1	73/73	100.0	0	0/247
Synthetic	B2	73/73	100.0	0	0/247
Synthetic	M1	73/73	100.0	0	0/247
Synthetic	B3	73/73	100.0	0	0/247
WESAD	B0	0/71	0.0	71	0/336
WESAD	B1	74/74	100.0	0	0/349
WESAD	B2	74/74	100.0	0	0/349
WESAD	M1	74/74	100.0	0	0/349
WESAD	B3	73/73	100.0	0	0/348
PPG-DaLiA	B0	0/80	0.0	80	0/326
PPG-DaLiA	B1	80/80	100.0	0	0/329
PPG-DaLiA	B2	80/80	100.0	0	0/330
PPG-DaLiA	M1	80/80	100.0	0	0/328
PPG-DaLiA	B3	80/80	100.0	0	0/330

Table S14. Original paired descriptive method contrasts with the original uncertainty procedure.

Source	Contrast	Outcome	N (S/E)	Difference (pp)	95% interval (pp)
Synthetic	M1-B2	Correction	10/20	0.00	[0.00, 0.00]
Synthetic	M1-B1	Case error	10/20	0.00	[0.00, 0.00]
Synthetic	M1-B1	Fact recall	10/20	0.00	[0.00, 0.00]
WESAD	M1-B2	Correction	10/20	0.00	[0.00, 0.00]
WESAD	M1-B1	Case error	10/20	0.00	[0.00, 0.00]
WESAD	M1-B1	Fact recall	10/20	0.00	[0.00, 0.00]
PPG-DaLiA	M1-B2	Correction	10/20	0.00	[0.00, 0.00]
PPG-DaLiA	M1-B1	Case error	10/20	0.00	[0.00, 0.00]
PPG-DaLiA	M1-B1	Fact recall	10/20	-0.25	[-0.75, 0.00]

Table S15. Original three-claim symbolic fixture, kept separate from the new 240-case structural programs.

Method	Corrected/required	Residual	Collateral/unaffected
B0	0/4	4	0/12
B1	2/4	2	0/12
B2	2/4	2	0/12
M1	4/4	0	0/12
B3	4/4	0	0/12

Table S16. Original hardware/model settings and sampled resource peaks. Sampling can miss brief peaks.

Component	Recorded setting
GPU	RTX 5090; 31.36 GiB
Primary model	Qwen3-8B; BF16; CUDA only
Model revision	b968826d9c46
Inference	vLLM 0.10.2; PyTorch 2.8.0+cu128
Database	Neo4j 5.26.0; SQLite 3.45.1 WAL
Input / output limit	4,096 / 1,536 tokens
Batch concurrency / interactive	4 / 1
CPU quota / memory limit	27.2 CPU equivalents / 94.0 GB
Test GPU memory peak	28,441.0 MiB
Test GPU utilization mean	88.7%
Test host cgroup memory peak	28.5 GiB

Table S17. Original failures, bounded repairs and supported references. Three B0 transport errors remain recorded failures.

Source	Method	Failed	Truncated	Repairs	Abstain%	Valid refs%
Synthetic	B0	1	0	0	0.4	48.9
Synthetic	B1	0	0	127	8.3	100.0
Synthetic	B2	0	0	127	8.3	100.0
Synthetic	M1	0	0	127	8.3	100.0
Synthetic	B3	0	0	127	8.3	100.0
WESAD	B0	1	0	0	0.4	52.2
WESAD	B1	0	0	141	8.3	100.0
WESAD	B2	0	0	140	8.3	100.0
WESAD	M1	0	0	141	8.3	100.0
WESAD	B3	0	0	142	8.3	100.0
PPG-DaLiA	B0	1	0	0	0.4	52.9
PPG-DaLiA	B1	0	0	137	8.3	100.0
PPG-DaLiA	B2	0	0	136	8.3	100.0
PPG-DaLiA	M1	0	0	138	8.3	100.0
PPG-DaLiA	B3	0	0	137	8.3	100.0

Table S18. Original narrow background-scope audit. These facts remain invalid under the frozen target-only contract.

Source	B0 contract-invalid claims	Locally supported off-target facts
Synthetic	254	107
WESAD	278	115
PPG-DaLiA	267	111

Table S19. Original graph versus relational generated-outcome contrasts. Logical parity under common candidates is measured separately in the new extension.

Source	Outcome	N (S/E)	M1-B3 (pp)	95% interval (pp)
Synthetic	Case error	10/20	0.00	[0.00, 0.00]
Synthetic	Fact recall	10/20	0.00	[0.00, 0.00]
Synthetic	Correction	10/20	0.00	[0.00, 0.00]
WESAD	Case error	10/20	0.00	[0.00, 0.00]
WESAD	Fact recall	10/20	0.25	[0.00, 0.75]
WESAD	Correction	10/20	0.00	[0.00, 0.00]
PPG-DaLiA	Case error	10/20	0.00	[0.00, 0.00]
PPG-DaLiA	Fact recall	10/20	-0.25	[-0.75, 0.00]
PPG-DaLiA	Correction	10/20	0.00	[0.00, 0.00]

Table S20. Original systems events, backlog and latency accounting. Finite candidate-completion latency does not represent adequate replacement time.

Events/s	Method	Flag p95 (ms)	Candidate p95 (s)	Flag misses	Repair misses	Event Q	Answer Q
1	M1	43.9	11.0	0/2	2/2	0	2
1	B3	3.7	10.9	0/2	2/2	0	2
5	M1	52.8	18.7	0/8	8/8	0	7
5	B3	23.7	19.5	0/8	8/8	0	7
20	M1	33,687.0	218.7	21/30	30/30	347	62
20	B3	77,698.4	237.9	17/30	30/30	394	50

Table S21. Flagging and candidate completion versus adequate replacement: one 60-second cell per offered rate and method. All successful-replacement times remain unresolved.

Method	Events/s	Flag p95 ms	Candidate p95 s	Adequate
M1	1	43.9	11.0	0/5
B3	1	3.7	10.9	0/5
B3	5	23.7	19.5	0/23
M1	5	52.8	18.7	0/23
M1	20	33687.0	218.7	0/90
B3	20	77698.4	237.9	0/90

Table S22. Completion-time adequacy audit. Zero adequate answers makes the zero became-stale count uninformative about robust recovery.

Events/s	Method	Calls	Input adequate	Completion adequate	Became stale
1	M1	5	0	0	0
1	B3	5	0	0	0
5	B3	23	0	0	0
5	M1	23	0	0	0
20	M1	90	0	0	0
20	B3	90	0	0	0

Table S23. Original request/token accounting by workload. Extension costs appear separately in Table S8.

Workload	Requests	No usage	Input tokens	Output tokens	Wall min
Matched test	5,220	3	8,525,358	2,880,024	173.2
pilot v1	207	0	256,281	103,092	11.4
pilot v2	164	0	230,554	90,664	9.6
pilot v3	186	0	300,657	106,302	9.3
pilot v4	187	0	302,546	106,911	9.3
Systems test	472	0	416,220	206,856	15.1
Automated audit	64	0	143,213	2,786	0.6
Systems pilot v1	8	0	6,792	3,060	0.4
Systems pilot v2	8	0	6,792	3,060	0.4

S5 Reproduction and Artifact Map

From the project root, the existing virtual environment and downloaded prepared events suffice for CPU analysis; no minimum-v1 model calls are repeated. The commands and their complete options are maintained in `docs/semantic_reproduction.md`. Run the semantic modules in order: analysis, structural analysis, alias analysis, figures and publication. Their exact module names and commands are listed in that document. Then run `bash paper/build.sh`. Focused tests and the PDF/integrity checker are included.

Actual original database exports reside in `artifacts/analysis/semantic_analysis_v1/exports/`; the matching manifest preserves Cypher, scope, export time, checksums and representative stored paths. Selected structural database exports live under `structure_exports/`. Semantic projection sidecars and figure layout files retain full IDs and derived-path witnesses. The original inputs and runtime hashes are verified against the minimum-run manifest. New inference uses `artifacts/runs/structure_study_v1/`, with core prompts, raw responses, repairs, CUDA placement checks, request journals, resource samples and all method/update logs. The separate normalization replication has its own manifest, candidates and replay directory.

This Supplement and the main manuscript use the official LLNCS template. The final report records the actual page counts, figure checks and source/result hashes. No conference submission is performed by the reproduction workflow.